FIRST SEMESTER M.Sc. DEGREE EXAMINATION, DECEMBER 2019

(CUCSS)

Physics

PHY IC 01—CLASSICAL MECHANICS

(2017 Admissions)

Time : Three Hours

Maximum : 36 Weightage

Section A

Answer **all** *questions.*

Each question carries weightage of 1.

1. Obtain the equation of motion of simple pendulum using Lagrangian method
2. Explain D' Alembert's principle.
3. What is the condition for a transformation to be canonical ?
4. What are canonical transformations ? What is its importance ?
5. Explain the physical significance of Hamilton's principal function.
6. Show that rigid bodies have six degrees of freedom
7. Define poisson's bracket.
8. What is coriolis force ?
9. What are normal vibrations and normal co-ordinates ?
10. Write any two examples for stable equilibrium.
11. What is chaos ?
12. What is bifurcation ?

(12 × 1 = 12 weightage)

Section B

Answer any **two** *questions.*

Each question carries weightage of 6.

13. What is Δ - variation ? How it differ from δ - variation ? State and prove the principle of least action
14. State Hamilton-Jacobi equation for Hamiltons principal function and explain how it can be used to solve Kepler's problem for a particle in an inverse square central force field.

Turn over

15. (a) Deduce Euler's equation of for the motion of rigid body.

(b) Find the kinetic energy of rotation of a rigid body with respect to principal axes in terms of Euler's angles and interpret the results when $I_1 = I_2$.

16. Using the logistic map, explain period doubling, bifurcation, Feigenbaum constants intermittency and chaos.

(2 × 6 = 12 weightage)

Section C

Answer any **four** *questions.*
Each question carries weightage of 3.

17. Derive Lagrange's equation from Hamilton's principle.

18. Show that $P = 2\left(1+q^{\frac{1}{2}}\cos p\right)q^{\frac{1}{2}}\sin p$ and $Q = \log\left(1+q^{\frac{1}{2}}\cos p\right)$ is canonical.

19. What is the effect of coriolis force on a particle falling freely under the action of gravity ?

20. Two masses m and 3m are attached to the two ends of a massless spring with force constant, k. Find the natural angular frequency of oscillation, if m = 100g and k = 0.3 N/m.

21. Three particles of equal mass (m) are connected by two identical massless spring constant (k). Find the potential energies of the system, if x_1, x_2 and x_3 denote the horizontal displacement of the masses from their respective equilibrium positions.

22. Outline the fixed point analysis of the damped linear harmonic oscillator.

(4 × 3 = 12 weightage)